

Enhancing Accuracy and Interpretability in Corporate Credit Rating Classification with the Transformer-LSTM Model.

Kensei Monden^a, Suguru Yamanaka¹

^aGraduate School of Science and Engineering, Aoyama Gakuin University, 5-10-1 Fuchinobe Chuo-ku, Sagami-hara-shi, 252-5258, Kanagawa, Japan

Abstract

Several machine learning methods have been applied to corporate credit rating classification, aimed at assessing a corporation's ability to repay debts based on it. Previous research has demonstrated that neural network models exhibit high accuracy in classification tasks, particularly when bolstered by time-series features. However, the challenge with neural networks lies in their interpretability due to their nonlinear and intricate calculation processes. In light of this, we introduce a novel approach known as a Transformer-long short-term memory (T-LSTM). Our goal is to strike a balance between classifier interpretability and the utilization of time-series features. Subsequently, we apply the T-LSTM model to tackle the corporate credit rating classification problem. The outcomes of our comparative experiments illustrate the superiority of T-LSTM in capturing time-series features when contrasted with conventional machine learning methods. Moreover, we analyze T-LSTM parameters to elucidate the point at which the acquired information proves beneficial for classification. Moreover, through an analysis of T-LSTM parameters, we reveal how it effectively leverages specific temporal attributes contingent upon the corporate credit rating being predicted. Thus, we present compelling evidence that T-LSTM enhances both the accuracy and interpretability of corporate credit rating classifications.

Keywords:

Attention mechanism, Corporate credit rating classification, LSTM, Recurrent neural networks, Transformer

1. Introduction

Corporate credit rating classification, aimed at assessing a company's repayment capability using financial statements, has embraced mathematical models to streamline the evaluation process. In recent years, machine learning-based mathematical models have gained prominence in credit rating classification [1–4]. Previous research has demonstrated that neural networks yield impressive accuracy in classification tasks [5–7], and has employed the neural network model for a benchmark classifier [1–4]. However, these models are often challenging to interpret due to their nonlinear and intricate calculation processes. This interpretability issue holds significant importance for practical business applications. In this study, we introduce the Transformer-long short-term memory (T-LSTM), a credit rating classifier that combines Transformer and LSTM compo-

nents. What sets T-LSTM apart is its interpretability, characterized by transparent parameters, and its ability to effectively capture time-series features. Our results confirm the superiority of T-LSTM over conventional machine learning models, underscoring the value of time-series features in enhancing classification accuracy. Furthermore, our parameter analysis reveals the pivotal financial statements at each time step that contribute to classification, providing valuable insights for decision-making.

Corporate credit risk assessment plays a pivotal role in investment decision-making and financial risk management. Typically, credit analysts rely on a combination of quantitative and qualitative corporate attributes, including financial statements, to determine corporate credit ratings. To streamline this evaluation process, various statistical models and machine learning methods are employed. These encompass techniques such as logistic regression [8], support vector machines (SVM) [2, 9, 10], neural networks, random forest (RF) [3], gradient boosting tree methods like XGBoost [11], and the

Email address: syamanaka@math.aoyama.ac.jp (Suguru Yamanaka)

least-squares probabilistic classifier (LSPC) [12].

Most prior research has traditionally relied on recently observed financial statements as input data for corporate credit rating classification. However, it is important to note that a company's value and repayment capability are significantly influenced not only by current financial conditions but also by historical financial time series data. Consequently, some recent studies have adopted time-series data from financial statements as input to leverage the valuable time-series features. Examples of previous studies that have utilized time-series data as input for corporate credit rating classification include references [13] and [14]. These studies employed recurrent neural networks (RNNs), which possess structures well-suited for handling time-series data, as predictive models. They concluded that incorporating time-series features significantly enhances classification accuracy. Specifically, [13] utilized gated recurrent units (GRU) [15], while [14] employed the long short-term memory model (LSTM) [16].

While neural network models exhibit outstanding performance, it is important to note that their calculation process is inherently nonlinear and complex, which can pose certain challenges. This issue becomes particularly critical in real-world applications where the rationale behind specific decisions or predictions is essential. To tackle this challenge, various methods have been introduced, focusing on enhancing the interpretability and transparency of the model. Sharp values, a concept borrowed from game theory to represent the contributions of participants in a game, are employed to assess the contribution and significance of the feature values within the input data [17]. A global surrogate model, which is an interpretable model trained to predict uninterpretable models such as neural networks, is also adopted [18].

In this study, we propose T-LSTM, taking into account classifier interpretability and leveraging time series features. The Transformer, a state-of-the-art technique employed in natural language processing [19], plays a pivotal role in our approach. This method correlates multiple time-steps to capture useful features and representations. Recent studies have suggested that the performance of this method is extremely high in both natural language processing and various fields [20–22]. Furthermore, by analyzing the correlations, we can confirm which financial statements at each time step are important or useful in making decisions and predictions. In this regard, the transformer is considered an interpretable method. Therefore, we employed this method to improve the classifier's interpretability. We also employed LSTM to capture time-series features to improve

classification accuracy. By employing these methodologies, we can simultaneously address both classifier interpretability and the incorporation of time-series features into corporate credit rating assessments.

The remainder of this paper is organized as follows. Section 2 presents the formulation of the corporate credit rating classification task and describes the T-LSTM. Section 3 details the dataset, encompassing data preprocessing steps, as well as the baseline models employed for comparison. Section 4 showcases the experimental results and conducts an analysis of the temporal dependencies learned by the transformer. Finally, Section 5 concludes the paper.

2. Proposed Method

In this section, we introduce the formularization of corporate credit rating classification and T-LSTM, which is based on a Transformer and LSTM.

The credit rating classification model for financial time-series data aims to establish a function wherein the inputs consist of financial time-series data, and the outputs correspond to credit ratings. Thus, given financial-statements time-series data $\mathbf{X}_i^{T,\ell} = \{\mathbf{x}_i^{T-\ell+1}, \dots, \mathbf{x}_i^T\} (\mathbf{x}_i^{T-\ell+1} \in \mathbb{R}^n)$ as input data, where i is an index of the corporate enterprise and n is the number of the financial indicators, we formulate the corporate credit rating classification as an estimation of the function $Y_i^T = F(\mathbf{X}_i^{T,\ell})$. $T \in \mathbb{N}$ and $\ell \in \{1, 2, \dots, T\}$ denote the time steps required to evaluate credit ratings and the length of the time-series data, respectively. $Y_i^T \in \{1, \dots, K\}$ denotes the predicted credit rating for input $\mathbf{X}_i^{T,\ell}$.

In this study, we propose a T-LSTM consisting of a Transformer and LSTM (Fig.1) as the function F .

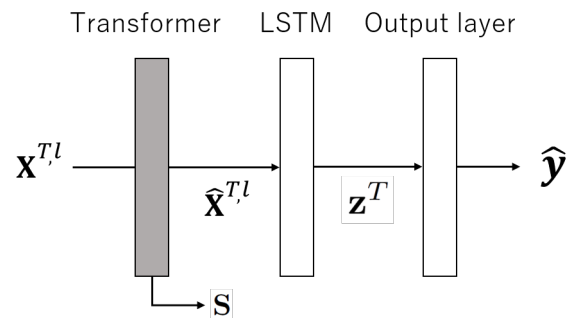


Fig. 1. Structure of the T-LSTM.

To obtain interpretable parameters, we assigned a transformer to the first layer of our model. In our model,

each time step of the time-series data is correlated with the transformer, and we can analyze the correlations among the inputs to explain why certain predictions are made.

First, we generate three matrices called query, key, and value, from $\mathbf{X}_i^{T,\ell} = (\mathbf{x}_i^{T-\ell+1}, \dots, \mathbf{x}_i^T)^\top \in \mathbb{R}^{\ell \times n}$ (where $\top$ for transpose) as follows:

$$\mathbf{Q} = \mathbf{X}_i^{T,\ell} \mathbf{W}_q \quad \mathbf{K} = \mathbf{X}_i^{T,\ell} \mathbf{W}_k \quad \mathbf{V} = \mathbf{X}_i^{T,\ell} \mathbf{W}_v, \quad (1)$$

where $\mathbf{W}_{(q,k,v)} \in \mathbb{R}^{n \times n}$ represent weight matrices. In summary, $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ can be conveniently generated by weighting $\mathbf{X}_i^{T,\ell}$. The three matrices consist of time-step features, which are then correlated as follows:

$$\mathbf{S} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{n}}\right), \quad (2)$$

where the softmax function is given by:

$$\text{softmax}(u_k) = \frac{\exp(u_k)}{\sum_{k'=1}^K \exp(u_{k'})}. \quad (3)$$

where class label k refers to the credit rating $\{1, 2, \dots, K\}$. The softmax function was applied along the rows of $\mathbf{Q}\mathbf{K}^\top$. The generated matrix $\mathbf{S}$ indicates the relationship between time steps. To confirm this, we illustrate the calculation process of $\mathbf{Q}\mathbf{K}^\top$ as Fig.2. Due to the method used for generating $\mathbf{Q}$ and $\mathbf{K}$, each block denoted by $\boxed{t}$ in Fig.2 represents the features at time step t . The (j, i) element of $\mathbf{Q}\mathbf{K}^\top$ corresponds to the inner product of the features at time steps j and i . Therefore, element (j, i) signifies the relationship between time steps j and i . Then, $\mathbf{S}$ is generated by comparing each time step and applying the softmax function. Thus, we gain insight into which time steps hold significance or utility in influencing decisions and predictions through the analysis of $\mathbf{S}$. We employ this matrix to weight $\mathbf{V}$ in the following manner:

$$\tilde{\mathbf{X}}_i^{T,\ell} = \mathbf{S}\mathbf{V}. \quad (4)$$

Then, residual connections [23] and nonlinear transforms are applied as follows:

$$\hat{\mathbf{X}}_i^{T,\ell} = \tanh(\mathbf{X}_i^{T,\ell} + \tilde{\mathbf{X}}_i^{T,\ell} + \text{MLP}(\mathbf{X}_i^{T,\ell} + \tilde{\mathbf{X}}_i^{T,\ell})), \quad (5)$$

Here, the term "MLP" represents a non-linear transformation performed by a multilayer perceptron, which alters the number of units in the following sequence: $n \rightarrow 4n \rightarrow n$ through a single hidden layer of size $4n$ activated by ReLU. This transformation has been recommended for enhancing the data feature representations [19]. As a result, we derive $\hat{\mathbf{X}}_i^{T,\ell}$ as the output of the

Transformer. $\hat{\mathbf{X}}_i^{T,\ell}$ is interpreted as time-series data, taking into consideration the relationships among all time steps.

Second, we apply LSTM to $\hat{\mathbf{X}}_i^{T,\ell}$:

$$\mathbf{z}_i^T = \text{LSTM}(\hat{\mathbf{X}}_i^{T,\ell}). \quad (6)$$

LSTM, as introduced in [16], falls under the category of RNNs. It is explicitly designed to capture long-term dependencies from time-series data and comprises a distinct unit known as a memory unit. A defining characteristic of LSTM is its incorporation of three gates—namely, the input gate, forget gate, and output gate—to govern information processing at each time step. These gates are responsible for managing and regulating the necessary information. We elucidate the LSTM process at time step t in Fig.3, where $\mathbf{x}^t \in \mathbb{R}^n$ and $\mathbf{z}^t \in \mathbb{R}^h$ represent the input and output data at time step t , respectively. Here, h , referred to as the number of units, is a hyperparameter determined based on the input data size n . $\mathbf{s}$ denotes the memory cell responsible for retaining the unit's state. The input gate, denoted as $\mathbf{i}$, controls the information to be incorporated. The output gate, denoted as $\mathbf{o}$, specifies the information to be utilized as the output. The forget gate, represented by $\mathbf{f}$, dictates the information to be discarded. $\mathbf{u}$ signifies the unit that receives and activates input data, while A stands for the unit responsible for activating the memory cell's output $\mathbf{s}$.

At every time step t , LSTM presents the input $\mathbf{x}^t$ at the present time step and the state $\mathbf{s}^{t-1}$ and output $\mathbf{z}^{t-1}$ at the previous time step. Then, it computes the state $\mathbf{s}^t$ and the output $\mathbf{z}^t$ at the present time step. Given $\hat{\mathbf{X}}_i^{T,\ell} = \{\hat{\mathbf{x}}_i^{T-\ell+1}, \dots, \hat{\mathbf{x}}_i^T\} (\hat{\mathbf{x}}_i^{T-\ell+1} \in \mathbb{R}^n)$ which is the output of Transformer, the calculation process at time step t is formulated as follows:

In the first step, LSTM weighs and activates the input $\hat{\mathbf{x}}_i^t$ to generate $\tilde{\mathbf{s}}_i^t$.

$$\tilde{\mathbf{s}}_i^t = \tanh(W_x^s \hat{\mathbf{x}}_i^t + W_z^s \mathbf{z}_i^{t-1}). \quad (7)$$

In the second step, LSTM generates the value $\mathbf{i}_i^t$ of the input gate to determine the information that should be added.

$$\mathbf{i}_i^t = \text{sigmoid}(W_x^i \hat{\mathbf{x}}_i^t + W_z^i \mathbf{z}_i^{t-1} + W_s^i \mathbf{s}_i^{t-1}) \quad (8)$$

where the sigmoid function is given by:

$$\text{sigmoid}(u) = \frac{1}{1 + \exp(u)}, \quad u \in \mathbb{R}. \quad (9)$$

LSTM also generates the value $\mathbf{f}_i^t$ of the forget gate to determine the information that should be removed.

$$\mathbf{f}_i^t = \text{sigmoid}(W_x^f \hat{\mathbf{x}}_i^t + W_z^f \mathbf{z}_i^{t-1} + W_s^f \mathbf{s}_i^{t-1}). \quad (10)$$

$$\mathbf{Q}\mathbf{K}^\top = \begin{pmatrix} t=1 \\ t=2 \\ t=3 \\ \vdots \\ t=T \end{pmatrix} \begin{pmatrix} t \parallel 1 & t \parallel 2 & t \parallel 3 & \cdots & t \parallel T \end{pmatrix} = \begin{pmatrix} t=1 & t=1 & t=1 & t=2 & t=1 & t=3 & \cdots & t=1 & t=T \\ t=2 & t=1 & t=2 & t=2 & t=2 & t=3 & \cdots & t=2 & t=T \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ t=T & t=1 & t=T & t=2 & t=T & t=3 & \cdots & t=T & t=T \end{pmatrix}$$

Fig. 2. Calculation process of $\mathbf{Q}\mathbf{K}^\top$.

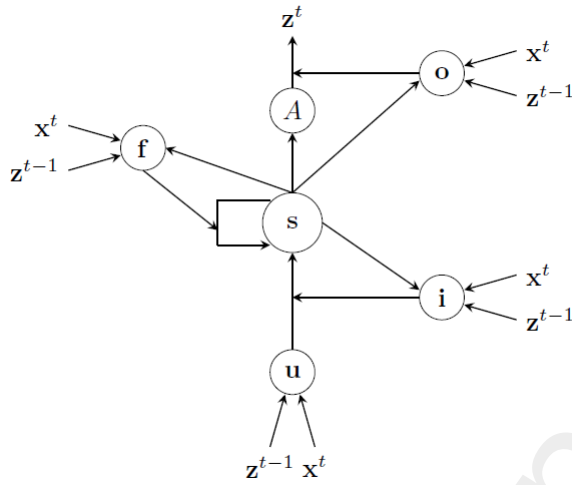


Fig. 3. LSTM structure.

In the third step, the new state value $\mathbf{s}_i^t$ is calculated based on the values generated above.

$$\mathbf{s}_i^t = \mathbf{f}_i^t \circ \mathbf{s}_i^{t-1} + \mathbf{i}_i^t \circ \tilde{\mathbf{s}}_i^t, \quad (11)$$

where $\circ$ denotes Hadamard product. Then, LSTM generates the value $\mathbf{o}_i^t$ of the output gate to determine the information that should be output.

$$\mathbf{o}_i^t = \text{sigmoid}(W_x^o \mathbf{x}_i^t + W_z^o \mathbf{z}_i^{t-1} + W_s^o \mathbf{s}_i^t). \quad (12)$$

In the last step, the output $\mathbf{z}_i^t$ is calculated as

$$\mathbf{z}_i^t = \mathbf{o}_i^t \circ \tanh(\mathbf{s}_i^t). \quad (13)$$

Note that W_v^u ($u \in \{\tilde{s}, i, f, o\}, v \in \{x, z, s\}$) are weight matrices. By applying the aforementioned process, we generate the vector $\mathbf{z}_i^T \in \mathbb{R}^h$ which is the final representation that considers all time-series features.

Finally, we transform vector $\mathbf{z}_i^T \in \mathbb{R}^h$ into the following predictions:

$$\hat{\mathbf{y}}_i = \text{softmax}(W\mathbf{z}_i^T + \mathbf{b}), \quad (14)$$

where $W \in \mathbb{R}^{K \times h}$ and $\mathbf{b} \in \mathbb{R}^K$ are the weight matrix and bias vector, respectively. K represents the number of predicted classes. The class predicted $\hat{Y}_i$ by the T-LSTM is calculated as

$$\hat{Y}_i = \arg \max_{k \in \{1, \dots, K\}} (\hat{y}_{k,i}). \quad (15)$$

Here, $\hat{y}_{k,i}$ is the k -th element of $\hat{\mathbf{y}}_i$. Currently, we have completed the formulation of our T-LSTM credit rating classification model.

3. Experimental Setup

In this section, we provide an overview of the dataset and outline the data preprocessing steps employed for credit rating classification utilizing T-LSTM. Additionally, we detail the configuration of the benchmark models and the methodologies used to assess the performance of each classification method.

3.1. Dataset

We compiled a dataset by collecting data from publicly disclosed corporate financial statements spanning the years 1999 to 2019. This dataset included credit ratings assigned by Rating and Investment Information, Inc., one of Japan's largest credit rating agencies. The original credit rating grades covered a range from AAA to D. For the purposes of this study, we grouped these grades into three distinct class labels, as outlined in Table1. Our objective was to predict these class labels. Table1 also provides details on the sample sizes within each class. To construct the input features for our credit rating classification model, we selected 76 specific financial indicators, as listed in Table2.

Since the selected financial indicators originally span various scales, we apply standardization as a feature-scaling technique. This standardization process is executed as follows:

$$x_{i,n}^t = \frac{x_{i,n}^t - \mu_n^t}{\sigma_n^t}, \quad (16)$$

Table 1: Mapping of credit ratings to labels.

Credit ratings	AAA ~ A+	A, A-	BBB+ ~ D
Class labels	0	1	2
Size of samples	1035	1474	835

where $\{x'_{i,n}\}$ ($n = 1, 2, \dots, 76$) denotes the values of the data before standardization. Parameters μ'_n and σ'_n represent the mean and variance of $x'_{i,n}$ respectively. We then use the values applied to the standardization as inputs. If there were missing values in the data, the means of the previous time-step values were substituted for the missing values. We then remove records that still have missing values after substitution.

3.2. Implementation of the T-LSTM and baseline models

T-LSTM, the model introduced in this study, is comprised of both a Transformer and an LSTM. The computational process of T-LSTM has been elucidated in Section 2. As recommended in previous studies [13, 14], we set the number of units to $h = 150$. For training, we utilize the Adam optimizer [24] and implement early stopping with validation data.

In our evaluation, we compared the classification performance of T-LSTM to the typical benchmark models employed in the credit rating classification literature: long-short-term memory (LSTM), support vector machine (SVM), random forest (RF), and light gradient boosting machine (LGBM). To assess the efficacy of integrating the Transformer component with LSTM in our T-LSTM model, we include a regular LSTM in the benchmark models. This LSTM model also leverages time-series features, and we maintain the number of units at 150, consistent with the T-LSTM configuration. Furthermore, we employ the Adam optimizer for training this ordinary LSTM. A support vector machine (SVM) is a traditional machine learning method used in discriminant analysis. SVM has often been applied to corporate credit-rating classifications [3, 9, 10]. Generally, we must use the strategy “One Against One” [25] or “One Against All” [26, 27] to apply SVM to multi-class discriminant owing to SVM’s design to discriminate binary classes. In this study, we use the strategy “One Against All,” based on the results of [26]. The RBF kernel is employed as the kernel of SVM:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2), \quad (17)$$

where the value of γ and the coefficient of the normalization term are chosen as 0.3 and 32, respectively, by

grid search based on 5-fold cross-validation. Random forest (RF) is a machine learning method that conducts ensemble learning using decision trees. In this study, the number of decision trees and the depth of the trees are set to 1000 and 2048, respectively, by Grid search based on 5-fold cross-validation. Furthermore, we incorporate a gradient boosting tree method into our comparison. This approach incorporates a gradient descent algorithm into random forests, enhancing classification accuracy. In this study, we utilize the light gradient boosting machine (LGBM) [28], a member of the gradient boosting tree family. It is worth noting that we determined the hyperparameters, including the number of decision trees, tree depth, and the number of branches, through a grid search method based on 5-fold cross-validation. The final values selected were 300 decision trees, a tree depth of 10, and 130 branches.

We implement the above models using Python 3.9. We use the numpy and pandas packages for data preprocessing. In addition, we use Pytorch for T-LSTM and LSTM and sci-kit learn for SVM, RF, and LGBM.

3.3. Evaluation method

We evaluate the classification performance of all models for corporate credit rating classification using accuracy (AC) and macro-F1 ($F1_{macro}$). The accuracy is calculated as $AC = \frac{TP + TN}{TP + FP + FN + TN}$, where TP, TN, FP, and FN are the elements of the confusion matrix, as presented in Table 3. Accuracy is affected by the number of class samples [29]. Therefore, we employ macro-F1 ($F1_{macro}$) to evaluate the classification performance. $F1_{macro}$ [30] is calculated as follows:

$$F1_{macro} = \frac{2}{\frac{1}{PRE_{macro}} + \frac{1}{REC_{macro}}}. \quad (18)$$

Note that PRE_{macro} and REC_{macro} are defined as follows:

$$PRE_{macro} = \frac{\sum_k \frac{TP_k}{TP_k + FP_k}}{K} \quad REC_{macro} = \frac{\sum_k \frac{TP_k}{TP_k + FN_k}}{K},$$

where k and K are indices of each class and the number of classes, respectively.

4. Experiments and Discussion

In this section, we conduct a comparative analysis of the classification performance between our proposed model and the benchmark models. Additionally, we explore the relationships among time steps. Furthermore, we discuss the significance and relevance of financial statements at specific time steps in the context of determining credit ratings.

Table 2: List of selected financial indicators.

net sales	ordinary profit
total assets	owners' equity
cash flows from operating activities	net working capital
current ratio	asset ratio
current liabilities to cash flows from operating activities ratio	fixed liabilities to cash flows from operating activities ratio
interest-bearing liabilities to cash flows from operating activities ratio	debt-equity ratio
equity ratio	fixed assets to equity ratio
fixed assets to long-term capital ratio	interest coverage ratio
current liabilities to sales ratio	fixed liabilities to sales ratio
interest-bearing liabilities to sales ratio	change ratio to sales ratio
change ratio of total profit	change ratio of operating profit
change ratio of ordinary profit	change ratio of total assets
change ratio of owners' equity	gross profit margin
operating profit to sales ratio	ordinary profit to sales ratio
pretax income to sales ratio	administrative expense ratio
ordinary profit to total capital ratio	operating profit to total capital ratio
business profit to total capital ratio	ordinary profit to owners' equity ratio
operating profit to owners' equity ratio	cash flows from operating activities to sales ratio
total assets turnover	equity turnover
trade receivables turnover	inventories turnover
current assets turnover	fixed assets turnover
change rate of liabilities	change rate of interest-bearing liabilities
change rate of cash and deposits	change rate of fixed assets
cash and deposits ratio	discounted interest expense to sales ratio
total profit to discounted interest expense ratio	tangible fixed assets to total assets ratio
operating expenses to sales ratio	other income to sales ratio
other expenses to sales ratio	interest burden to sales ratio
ratio of current liabilities to cash flows from operating activities	ratio of current liabilities to owners' equity
total capital growth ratio	pretax net earnings for the period to total capital ratio
tangible fixed assets growth ratio	liabilities turnover
current liabilities turnover period	short-term borrowings turnover period
capital turnover period	fixed liabilities turnover period
days cash and deposits on hand	cash interval
current assets to total assets ratio	fixed assets to total assets ratio
inventories to total assets ratio	pretax net earnings for the period to total assets
logarithm of total assets	tangible fixed assets ratio
ratio of owners' equity to legal capital	financial leverage
tangible fixed assets turnover	net income growth ratio

Table 3: Confusion matrix.

Actual labels	Predicted labels		
		Positive	Negative
	Positive	TP	FP
	Negative	FN	TN

4.1. Classification Performance of Proposed T-LSTM model

Table 4 and Fig. 4, 5 show the classification performance of T-LSTM along with the baseline models on the test data. The length of the input time-series data is denoted as $\ell = 10$. The test data encompasses credit rating samples from the years 2018 and 2019. Meanwhile, the training data were drawn from credit rating samples spanning the years 1999 to 2017 and 1999 to 2018.

Table 4: Accuracy and $F1_{macro}$ obtained for the test data of 2018 and 2019.

	2018		2019	
	AC	$F1_{macro}$	AC	$F1_{macro}$
T-LSTM	0.9005	0.8420	0.9518	0.8989
LSTM	0.8969	0.8373	0.9119	0.8290
SVM	0.6635	0.6641	0.6850	0.6829
RF	0.7183	0.7221	0.7492	0.7489
LGBM	0.7404	0.7424	0.7217	0.7510

Table 4 and Figures 4 and 5 clearly illustrate that T-LSTM outperforms all other models in terms of both accuracy (AC) and $F1_{macro}$ for both test data sets. LSTM also exhibits superior performance compared to the remaining baseline models. Notably, what sets T-LSTM and LSTM apart from the other baseline models is their utilization of time-series information for credit rating determination. This underscores the critical role of time-series features in enhancing classification accuracy. Particularly noteworthy is the fact that T-LSTM surpasses the performance of ordinary LSTM. These results strongly suggest that the Transformer's ability to learn the importance of time steps significantly enhances LSTM's capability to capture time-series features, resulting in superior performance. Furthermore, the findings imply that the relationships between time steps can be distilled into a single critical value for analysis.

4.2. Analysis of the importance of time steps in time series features.

Fig. 6 presents the heat map of the relationships between time steps $\mathbf{S}$ calculated using equation (2) for the training data from 1999 to 2018 and test data of 2019. Index 1 and 10 denote the most past and recent time steps in Fig. 6. Each S_{ij} element of $\mathbf{S} \in \mathbf{R}^{10 \times 10}$ represents the contribution or influence of time step j at time step i . Indices j and i denote the source and target times, respectively, in Fig. 6. The importance of time step j , I_j ($1 \leq j \leq 10$), is defined as follows:

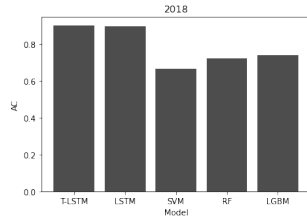
$$I_j = \sum_{i=1}^{10} S_{ij}. \quad (19)$$

The aggregated importance of the time steps for the training and test data are shown in Fig. 7. Fig. 7 shows that the financial statements on the first and last time steps are relatively important as compared to the middle time steps. We confirm that T-LSTM leverages the most past and recent time steps to determine the corporate credit ratings. Furthermore, we visualize the heat map of $\mathbf{S}$ and the importance of the time-steps the label in Fig. 8, 10 and Fig. 9, 11. Fig. 8(a) - 11(a) shows that the financial statements on time steps 1 and 10 are leveraged to determine label 0. However, we confirm that the financial statements at time steps 9 and 10, which are recent time steps, are leveraged to determine label 1 in Fig. 8(b) - 11(b). Compared with labels 0 and 2, all financial statements on the time steps were also utilized. Fig. 8(c) - 11(c) show that the model leverages the financial statements on time step 1 to determine label 2. The newest financial statements in time step 10 are as important as those in time step 1.

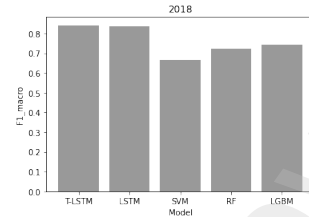
Therefore, we confirm that T-LSTM uses relationships and particular financial statements in time steps depending on the labels to be predicted. As previously mentioned, we can understand what the model utilizes to determine corporate credit ratings by analyzing its parameters.

5. Conclusion

In this study, we proposed T-LSTM, a model designed for corporate credit rating classification that offers interpretability while effectively capturing time-series features. Notably, T-LSTM's performance surpassed that of the baseline models, which included conventional machine learning models and LSTM. These findings highlight the value of incorporating time-series feature information, ultimately leading to improved classification accuracy. Furthermore, the Transformer

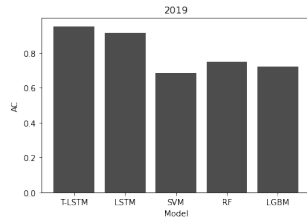


(a) Accuracy.

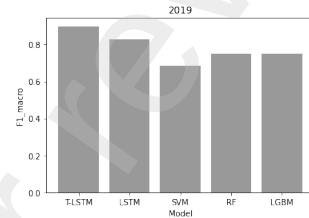


(b) $F1_{macro}$.

Fig. 4. Accuracy and $F1_{macro}$ for the data of 2018 in Table.4.

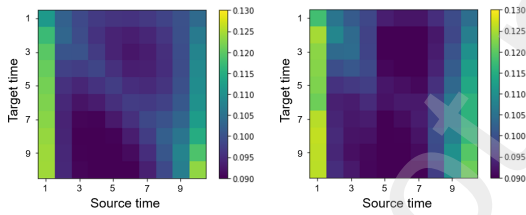


(a) Accuracy.



(b) $F1_{macro}$.

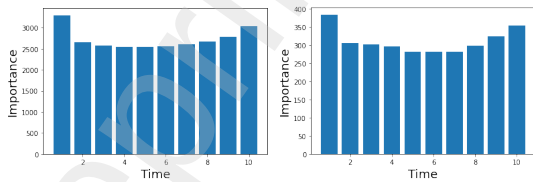
Fig. 5. Accuracy and $F1_{macro}$ for the data of 2019 in Table.4.



(a) training data.

(b) test data.

Fig. 6. Heat map of S for the training and test data.



(a) training data.

(b) test data.

Fig. 7. Aggregated time step importance $I_j (1 \leq j \leq 10)$ for the training and test data.

component's ability to learn the importance of time steps greatly enhances LSTM's efficiency in capturing time-series features. An in-depth analysis of the interpretable parameters within T-LSTM revealed that the model leverages specific financial statements at particular time steps, depending on the corporate credit rating being predicted. For instance, T-LSTM relies on financial statements from both the most recent and distant past time steps when determining labels 0 and 2. Conversely, for label 1, it predominantly utilizes financial statements from recent time steps. Based on the above analysis, we confirmed that the time-series features improved classification accuracy. This comprehensive analysis confirms the positive impact of time-series features on enhancing classification accuracy and sheds light on the decision-making process underlying corporate credit ratings. In essence, T-LSTM serves to improve both the accuracy and interpretability of corporate credit rating classification.

CRediT authorship contribution statement

Kensei Monden:Investigation, Methodology, Formal analysis, writing–original draft. **Suguru Yamana:**Conceptualization, data Curation, writing–review and editing, supervision, project administration, and funding acquisition.

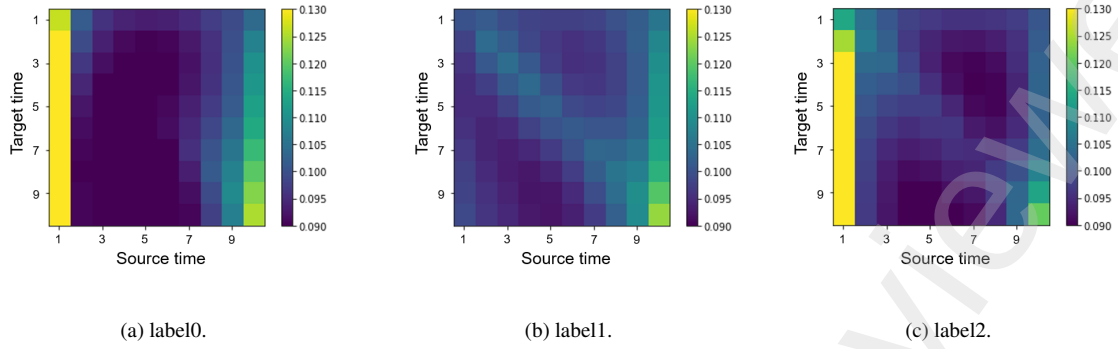


Fig. 8. Heat map of S for each label of the training data.

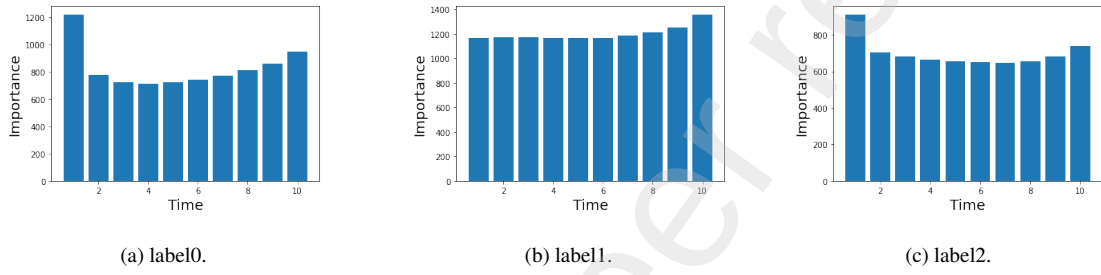


Fig. 9. Aggregated time step importance scores for the training data labels.

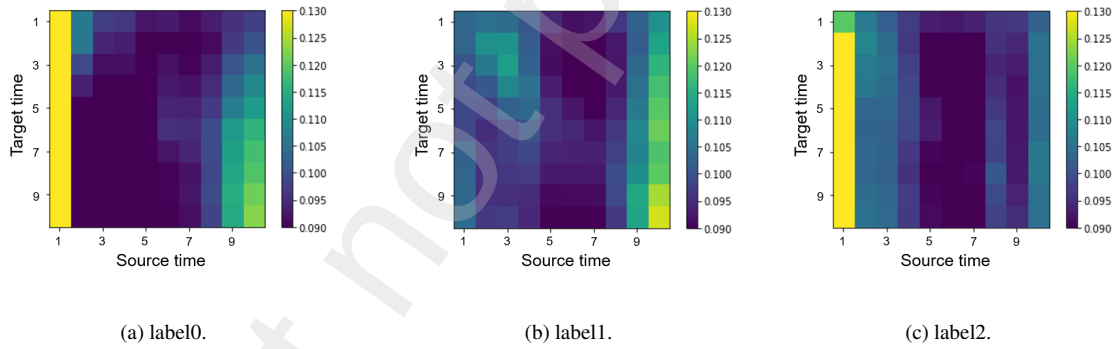


Fig. 10. Heat map of S for the test data labels.

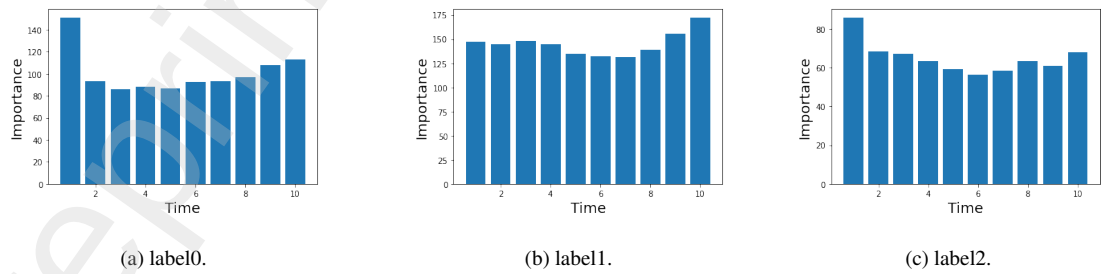


Fig. 11. Aggregated time step importance scores for test data labels.

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Declaration of Competing Interest

The authors declare that they have no competing financial interests or personal relationships that may have influenced the work reported in this study.

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